Type MS · Finding n from an equation linking sums and terms

Answer 1 of 11 · Question 1 part b · 5 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Number		
1 (a) (i)	$a+d+a+8d=0 \qquad a+3d+a+5d+a+9d=14$ Solve simultaneously $d=4$	M1 M1 A1
(ii)	$a=-18$	A1 (4)
(b)	$\frac{3n}{2}[48+6(n-1)]=\frac{2n}{2}[48+6(2n-1)]$ $3n(42+6n)=2n(42+12n) \Rightarrow 6n^2-42n=0$ $6n(n-7)=0 \Rightarrow n=[0,\ 7]$ $n=7$	M1 A1 A1 M1 A1 (5) **[9]**

Type MS · Finding n from an equation linking sums and terms

Answer 1 of 11 · Question 1 part b · 5 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For writing down both correct expressions in terms of a and d $a+d+a+8d=0$ $\quad$ $a+3d+a+5d+a+9d=14$
	M1	For attempting to solve their simultaneous equations for a and d $2a+9d=0$ $3a+17d=14$
	A1 (i)	For $d=4$ * **This is a show question – there must be no errors for the award of this mark**
	B1 (ii)	For $a=-18$ **This is an A mark in Epen**
(b)	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6\left[n-1\right]\right)=\frac{2n}{2}\left(2\times24+6\left[2n-1\right]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	For reaching a correct 2TQ equation in n $126n+18n^2=84n+24n^2\Rightarrow 6n^2-42n=0$
	M1	For attempting to solve **their** quadratic (See General Guidance for the definition of an attempt) $6n^2-42n=0\Rightarrow 6n\left(n-7\right)=0\Rightarrow n=\left[0,\ 7\right]$
	A1	$n=7$ Condone the value of 0 for this mark
	ALT	
	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6\left[n-1\right]\right)=\frac{2n}{2}\left(2\times24+6\left[2n-1\right]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	Divides through n to reach a linear equation to give $6n=42$ oe
	M1	Solves their linear equation in n
	A1	$n=7$

Type MS · Finding n from an equation linking sums and terms

Answer 1 of 11 · Question 1 part b · 5 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Paper 1		
Question number	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 2 of 11 · Question 6 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
6 (a)	$a = 4\times1-3=1, \quad (d=4)$ $\sum_{r=1}^{n} 4r-3 = \frac{n}{2}\left(2\times1+(n-1)4\right) = n(2n-1)^*$	B1 M1A1 [3]
(b)	$n(2n-1)>1000 \Rightarrow 2n^2-n-1000>0$ $\frac{-(-1)\pm\sqrt{(-1)^2-4\times2\times(-1000)}}{2\times2} \Rightarrow n>22.612... \Rightarrow n=23$	M1 M1A1 [3]
(c)	$3t_{(n+7)}+18 = S_{(n+4)}$ $\Rightarrow 3\left[4(n+7)-3\right]+18=(n+4)\left[2(n+4)-1\right]$ $\Rightarrow 2n^2+3n-65=0$ $2n^2+3n-65=(2n+13)(n-5)=0 \Rightarrow n=5$	 M1 A1 depM1A1 [4]
		Total 10 marks

Type MS · Finding n from an equation linking sums and terms

Answer 2 of 11 · Question 6 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 3 of 11 · Question 6 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1	
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1	
	$2\times17+d=38 \Rightarrow d=4$	A1	(4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1	(2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$		
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1	
	$2p^2-11p+12=0$	A1	
	$(2p-3)(p-4)=0 \Rightarrow p=4 \left(p\neq\frac{3}{2};\ \text{may not be seen}\right)$	M1A1	(4)
			[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $\left[a = S_1 = 1(15+2\times1) = 17\right]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}\left(2a+\left[n-1\right]d\right) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = \ldots$ and $30 = 2a-4 \Rightarrow a = \ldots$
		For the correct value of $d = 4$

Type MS · Finding n from an equation linking sums and terms

Answer 3 of 11 · Question 6 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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Type MS · Finding n from an equation linking sums and terms

Answer 3 of 11 · Question 6 part c · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$\left(\quad 9\right)^2 \qquad 81 \quad \left(\quad 9\right)^2 \quad 25$	

Type MS · Finding n from an equation linking sums and terms

Answer 4 of 11 · Question 8 part b · 4 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
8(a)	$42 = \frac{4}{2}\left(2a + (4-1)d\right) \Rightarrow 42 = 4a + 6d$ oe	B1
	$23 = a + 4d$	B1
	$\Rightarrow a = 3, \quad d = 5$	M1A1
	$n\text{th term} = 3 + (n-1)5 = 5n - 2 \Rightarrow S_n = \sum_{r=1}^{n}(5r-2)$	M1A1
	$[P = 5, \ Q = 2]$	[6]
(b)	$S_{2n} - 3U_n = 1062$	
	$\Rightarrow \frac{2n}{2}\left(2\times'3' + (2n-1)'5'\right) - 3\left['3' + (n-1)'5'\right] = 1062$	M1
	$\Rightarrow 10n^2 + n - 15n + 6 = 1062 \Rightarrow 10n^2 - 14n - 1056 = 0$	M1
	$10n^2 - 14n - 1056 = (5n + 48)(2n - 22) = 0$	M1
	$\Rightarrow n = 11$	A1
		[4]
		Total 10 marks

Type MS · Finding n from an equation linking sums and terms

Answer 4 of 11 · Question 8 part b · 4 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	Forms a correct equation in a and d for **either** the sum of the first 4 terms or for the 5th term This must be correct.
	B1	Forms correct equations in a and d for **both** the sum of the first 4 terms and for the 5th term Both must be correct.
	M1	Solves their two equations simultaneously by any method. Accept a pair of equations as follows: $pa+qd=21$ or 42 $ra+sd=23$ where $p\neq r$ and $q\neq s$ Allow a maximum of one arithmetical error in the solution of their SE.
	A1	For **both** $a=3$ and $d=5$
	M1	Attempts to form the required expression using their values of a and d They must use the nth term for this.
	A1	For the correct expression as written. This must be exactly as written in the question with the inclusion of the $\sum$. However, allow the omission of $S_n=.....$
	ALT for last 2 marks	
	M1	$S_n=\sum_{r=1}^{n}(Pr-Q)\Rightarrow 3=P-Q \quad 5=(2P-Q)-(P-Q)=P$ $\Rightarrow 3=5-Q\Rightarrow Q=2$
	A1	For the correct expression as written. This must be exactly as written in the question with the inclusion of the $\sum$. However, allow the omission of $S_n=.....$ Allow also $S_n=\sum_{r=1}^{n}(Pr-Q) \quad (\Rightarrow) \quad P=5, Q=2$ for this mark.
(b)	M1	Applies the summation formula and the nth term formula correctly to form an equation in n using their values for a and d For this mark you must see an equation, e.g., $\frac{2n}{2}(2\times 3+(2n-1)5)-3[3+(n-1)5]=1062$ However, allow recovery in the next step.
	M1	Forms a 3TQ $\quad 10n^2-14n-1056=0$ o.e. Condone missing = 0 if it is clear they are solving this equation.
	M1	Attempts to solve their 3TQ Accept any method. If their 3TQ is incorrect, only award this mark when you can see their method. If a calculator is used with an incorrect 3TQ and no method is seen, this mark is not available.

Type MS · Finding n from an equation linking sums and terms

Answer 4 of 11 · Question 8 part b · 4 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

		Note: We do not need to see $-\frac{48}{5}$. Some candidates will automatically reject this solution.
	A1	For the correct value of n Do not award if $-\frac{48}{5}$ is included as a value.

Question	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 5 of 11 · Question 6 part c · 5 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Mark
6(a)	$S_{20} = 20(3+2\times 20) = 860$	M1A1 [2]
(b)	$S_1 = 3\times 1+2\times 1^2 = 5$ $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$ $d = 9-5 = 4$ $U_n = '5'+(n-1)'4'$ $S_n = \sum_{r=1}^{n}(4r+1) \Rightarrow A = 4, \quad B = 1$	B1 M1 M1 A1 M1 A1 [6]
(c)	$T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right] \Rightarrow T_n = \frac{n}{2}(10+4n)$ or $T_n = n(5+2n)$ $n(5+2n) = 3n+2n^2+252, \quad 5n = 2n+252$ $5n = 3n+252 \Rightarrow n = 126$	M1 M1,dM1 ddM1A1 [5]
	Total	**13 marks**

Type MS · Finding n from an equation linking sums and terms

Answer 5 of 11 · Question 6 part c · 5 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
6(a)	M1	For substituting 20 into the given $S_{20} = 20(3+2\times 20) = 860$
	A1	For $S_{20} = 860$ Sight of 860 with no working scores M1A1
(b)	B1	For finding the first term $S_1 = 3\times 1+2\times 1^2 = 5$ or $a = 5$ Award this mark even if it is not clear that they understand that S_1 is the first term.
	M1	For a complete method to find the second term $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$
	M1	For finding the common difference, they must reach a value for this mark. $d = '9'-'5' = (4)$
	A1	For $d = 4$
	M1	For either $A = 4$ or $B = 1$ Accept embedded values.
	A1	For both $A = 4$ and $B = 1$ Accept embedded values.
	For the final correct answer seen without any or minimal working, award full marks in part (b)	
(c)	M1	For an expression for T_n using the given values $T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right]$
	M1	For equating their expression for T_n in $T_n = n(3+2n)+252$ The correct expression for S_n must be used here. $\Rightarrow \frac{n}{2}(10+4n) = n(3+2n)+252 \Rightarrow (5n = 3n+252)$ **This is an A mark in Epen**
	dM1	For forming a linear equation. e.g. $5n = 3n+252$ o.e. This mark is dependent on the previous M mark.
	ddM1	For solving **their** linear equation. $5n = 3n+252 \Rightarrow n = (126)$ This mark is dependent on both previous M marks.
	A1	For $n = 126$

Type MS · Finding n from an equation linking sums and terms

Answer 5 of 11 · Question 6 part c · 5 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Mark

Type MS · Finding n from an equation linking sums and terms

Answer 6 of 11 · Question 7 part c · 5 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
7 (a)(i)	$a+4d=4x+6$ $a+7d=7x+3$ $\Rightarrow 3d=3x-3\Rightarrow d=x-1$*	M1A1cso
(ii)	$a+7(x-1)=7x+3\Rightarrow a=10$ or $a+4(x-1)=4x+6\Rightarrow a=10$	M1A1 [4]
(b)	$42=10+8(x-1)\Rightarrow x=5$	M1A1 [2]
(c)	$d=x-1\Rightarrow d=5-1=4$	B1
	$S_{n+1}=12U_n+18\Rightarrow \frac{n+1}{2}\left(2\times 10+\left[(n+1)-1\right]4\right)=12\left[10+(n-1)4\right]+18$	M1
	$\Rightarrow n^2-18n-40=0$	M1
	$\Rightarrow (n-20)(n+2)=0\Rightarrow n=20$	M1A1 [5]
		Total 11 marks

Type MS · Finding n from an equation linking sums and terms

Answer 6 of 11 · Question 7 part c · 5 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 7 of 11 · Question 9 part c · 5 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

Number		
9 (a)	$a = 5\times1-1=4$ $d=5$ $S_n=\frac{n}{2}\left(2\times4+[n-1]5\right)=\frac{n}{2}(3+5n)*$ cso **ALT** $a=4$ $l=5n-1$ $S_n=\frac{n}{2}(a+l)=\frac{n}{2}(4+5n-1)=\frac{n}{2}(3+5n)*$ cso	B1 M1A1 [3] [B1 M1A1]
(b)	$\sum_{r=10}^{20}(5r-1)=\sum_{r=1}^{20}(5r-1)-\sum_{r=1}^{9}(5r-1)$ $\sum_{r=10}^{20}(5r-1)=\frac{20}{2}(3+5\times20)-\frac{9}{2}(3+5\times9)$ $=1030-216=814$ **ALT** $a=4+9\times5=49,\ l=4+20\times5=99,\ n=20-10+1=11$ $\sum_{r=10}^{20}(5r-1)=\frac{11}{2}(49+99)=814$ **ALT** $a=5\times10-1=49$ $d=5$ $n=11$ $S_n=\frac{11}{2}(2\times49+(11-1)\times5)=814$	B1 M1 A1 [3] [B1 M1A1] [B1 M1A1]
(c)	$\frac{n}{2}(3+5n)=12(4+5n)+52\Rightarrow5n^2-117n-200=0$ $\Rightarrow(n-25)(5n+8)=0\Rightarrow n=25,\ \left(n\neq-\frac{8}{5}\right)$	M1M1A1 M1A1 [5]
		Total 11 marks

Type MS · Finding n from an equation linking sums and terms

Answer 7 of 11 · Question 9 part c · 5 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

		May be implied by correct values seen in summation formula.
	M1	Uses a **correct** form of the summation formula for an arithmetic series with their a and their l provided their a and their l are stated. $S_n = \frac{n}{2}(a+l) = \frac{n}{2}('4' + '5n - 1')$
	A1 cso	For obtaining the given answer in full with no errors. $\sum_{r=1}^{n}(5r-1) = \frac{n}{2}(3+5n)$
Note: If standard summation results are correctly used award B1M1A1, if not fully correct send to review.		
(b)	**B1**	For correctly giving the required summation as the difference between two summations starting at $r = 1$. $\sum_{r=10}^{20}(5r-1) = \sum_{r=1}^{20}(5r-1) - \sum_{r=1}^{9}(5r-1)$
	M1	For substitution of $n = 20$ and $n = '9'$ into the result from part (a) and subtracting. $\frac{20}{2}(3+5\times 20) - \frac{'9'}{2}(3+5\times' 9')$ Allow for use of 9 or 10.
	A1	For the correct summation 814
Alternative method		
	B1	For finding the first term, last term and number of terms for the arithmetic sequence. $a = 49, l = 99, n = 11$
	M1	Uses a **correct** summation formula for an arithmetic series with their a, their $l$and their n provided these are stated. $\frac{'11'}{2}('49'+'99')$
	A1	For the correct summation 814
Alternative method – considering this as a series starting at the 10th term of the original series		
	B1	For finding the first term, common difference and number of terms. $a = 5\times 10 - 1 = 49$ $d = 5$ $n = 11$
	M1	Uses a **correct** form of the summation formula for an arithmetic series with their a, *their* d, and *their* n provided their a and their d and their n are stated and their $n \neq 20$ $S_n = \frac{'11'}{2}(2\times '49' + ('11'-1)\times '5')$
	A1	For the correct summation 814
(c)	**M1**	Uses $5r-1$ with $n+1$ to find an expression for u_{n+1} in terms of n. $5r-1 = 5(n+1)-1 = 5n+4$
	M1	Forms a correct equation for n using the result given in part (a) and their expression for $5r-1$ in terms of n. $\frac{n}{2}(3+5n) = 12('4+5n') + 52$
	A1	Obtains a correct 3TQ $5n^2 - 117n - 200 = 0$ oe
	M1	For an attempt to solve their 3TQ. See General Guidance for the definition of an attempt. $(n-25)(5n+8) = 0 \Rightarrow n = 25, \left(n = -\frac{8}{5}\right)$
	A1	For correct value: $n = 25$ If $n = -\frac{8}{5}$ is seen it must be rejected.

Type MS · Finding n from an equation linking sums and terms

Answer 7 of 11 · Question 9 part c · 5 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 1 of 10 · find part c in this mark scheme extract

Number		
2 (a)	$9(a+a+d)=a+4d+a+5d+a+6d$ oe $(5a=2d)$ oe $a+2d=12$ oe $5(12-2d)=2d$ oe $d=5 \quad a=2$	M1 B1 M1 A1 A1 [5]
For the main scheme, ALT1 and ALT2, give the first 1 or 2 M marks, regardless of if students carry on to use the method for the 3rd M mark. Favour the ALT which gives the most marks. **ALT3 may not be used except in the circumstances described below – clarify with your team leader if this is not clear.**		
(b)	$\sum_{n=1}^{60} u_n = \frac{60}{2}(2\times"2"+(60-1)\times"5")(=8970)$ $\sum_{n=1}^{14} u_n = \frac{14}{2}(2\times"2"+(14-1)\times"5")(=483)$ or $(u_{14})="2"+(14-1)\times"5"(=67)$ "8970" – "483" = 8487	M1 M1 ddM1 A1 [4]
ALT1	(First term = u_{15} =) $"2"+"5"\times(15-1)(=72)$ (Last term = u_{60} =) $"2"+"5"\times(60-1)(=297)$ or $"72"+"5"\times(46-1)(=297)$ $\frac{46}{2}("72"+"297")=8487$	M1 M1 ddM1 A1 [4]
ALT2	$(u_{14}=)"2"+"5"\times(14-1)(=67)$ $(u_{60}=)"2"+(60-1)\times"5"(=297)$ $\frac{60}{2}("2"+"297")-\frac{14}{2}("2"+"67")=8487$	M1 M1 ddM1 A1 [4]
This following ALT, may only be used if candidates are clearly and obviously going on to use $\left(\sum_{r=15}^{60} u_r = \right)\frac{46}{2}(2\times"72"+(46-1)\times"5")$ **Allow n = 45 for first 3 method marks.**		
ALT3	(First term =) $"2"+"5"\times(15-1)(=72)$ $\left(\sum_{r=15}^{60} u_r = \right)\frac{46}{2}(2\times"72"+(46-1)\times"5")$	M2 ddM1 A1 [4]
(c)	$2\times\frac{n}{2}(2\times2+(n-1)\times5)-5(5n-3)=10$ oe $(4n+5n^2-5n-25n+15=10)$ $5n^2-26n+5=0$ $(5n-1)(n-5)=0\rightarrow n=\frac{1}{5},5$ $n=5$	M1 A1 M1 A1 [4]
		Total 13 marks

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 2 of 10 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correctly forms the equation shown. Any equivalent unsimplified equation. Note: an alternative seen is (the mark is awarded for the second line below or any unsimplified equivalent) $(S_7 - S_4 = 9(a+a+d))$ $\frac{7}{2}(2a+(7-1)) - \frac{4}{2}(2a+(4-1)) = 9(a+a+d)$ $(\Rightarrow 15a = 6d)$
	B1	For the correct equation.
	M1	For a complete and valid method to solve the equations simultaneously, leading to either a = or d =. Allow one processing error **only.** Can be awarded for a complete and valid method to solve any two linear equations, both in a and d, one processing error only. Must lead to either a = or d =
	A1	For either the correct value of a or the correct value of d.
	A1	For the correct values of both a and d.
If candidates use a different method, follow the same principles to award marks or send to Review if this is not clear. Candidates may use any letters or symbols to denote a and d		
(b)	M1	Use of their a and d in the correct formula $\frac{n}{2}(2"a"+(n-1)"d")$ with $n = 60$
	M1	Use of their a and d in the correct formula $\frac{n}{2}(2"a"+(n-1)"d")$ with $n = 14$ Allow n = 15 as a concession.
	ddM1	For their $"\sum_{n=1}^{60} u_n" - "\sum_{n=1}^{14/15} u_n"$. Dependent on both previous method marks. Follow through their value from use of n = 15 if used. The presence of 8970 – 483 will imply M3 here as will 8487.
	A1	For 8487
ALT1	M1	For correct use of $"a"+(n-1)"d"$ with $n = 15$ with their a and their d
	M1	For correct use of $"a"+(n-1)"d"$ with $n = 60$ with their a and their d or correct use of $"a_{15}"+(n-1)"d"$ with $n = 46$ with their d and correctly using their term for a_{15} from the first M mark. Allow n = 45 as a concession.
	ddM1	For correct use of the formula $\frac{n}{2}(u_{15}+u_{60})$. Dependent on both previous method marks. Allow n = 45 as a concession. $\frac{46}{2}("72"+"297")$ or 8487 implies M3
	A1	For 8487
ALT2	M1	For correct use of $"a"+(n-1)"d"$ with $n = 14$ with their a and their d Allow n = 15
	M1	For correct use of $"a"+(n-1)"d"$ with $n = 60$ with their a and their d
	ddM1	For correct use of the formula $\frac{n}{2}(u_1+u_{60}) - \frac{n}{2}(u_{14}+u_{60})$. Dependent on both previous method marks. Allow n = 15 The presence of 8970 – 483 will imply M3 here as will 8487
	A1	For 8487

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 3 of 10 · find part c in this mark scheme extract

ALT3	M2	For correct use of $"a"+(n-1)"d"$ with $n=15$ with their a and their d
	ddM1	For correct use of the formula $\frac{n}{2}\left(2\times"u_{15}"+(n-1)"d"\right)$. Dependent on both previous method marks. Allow $n = 45$ as a concession.
	A1	8487
(c)	M1	For correct use of $2S_n-5(an+b)=10$ Substitution into S_n must be correct for their values of a and d. **and** candidates must correctly use their a in an expression of the form $an+b$ b can be any integer, $b\neq 0$ $2S_n-5(an+b)=10$ **does not have to be simplified to be awarded this mark.**
	A1	For the correct 3 term quadratic equation = 0. Equivalent coefficients are possible.
	M1	For any valid, complete method to solve their quadratic (see general guidance), which must be of the form $pn^2+qn+r \quad p,q,r\neq 0$ leading to $n=$ $n = 5$ will imply this mark.
	A1	$n = 5$ $n = 0.2$ must be dismissed.

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 4 of 10 · find part c in this mark scheme extract

Q	Scheme	Marks
3	$\frac{\sin 25}{7}=\frac{\sin B}{10}$ oe $(B=)\sin^{-1}\left(\frac{10\sin 25}{7}\right)$ oe $(B=37.138(254541)$ or $142.861(7455))$	M1dM1
	$(C=180-[(180-37.138(25454))+25])$	
	$(C=)12.138(25451)$	A1 (M1 on ePen)
	$(c^2=)7^2+10^2-2(7)(10)\cos"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{10}{\sin"142.861(7455)"}$ oe	
	eg $(c=)\frac{10}{\sin"142.861(7455)"}\times\sin"12.138(25451)"$	
	**allow students to follow through using their acute angle for B	ddM1
	$(c^2=)7^2+10^2-2(7)(10)\cos"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{10}{\sin"37.138(254541)"}$ oe	
	eg $(c=)\frac{10}{\sin"37.138(254541)"}\times\sin"117.861(7455)"$	
	$(c=)3.5$ (cm)	A1 [5]
ALT1	$7^2=10^2+(AB)^2-2(10)(AB)\cos 25$	M1
	$0=(AB)^2-20\cos 25(AB)+100-49$ $(20\cos 25=18.12615574)$	dM1
	$0=(AB)^2-20\cos 25(AB)+51$	A1 (M1 on ePen)
	$(AB=)\frac{"--20\cos 25"\pm\sqrt{("-20\cos 25")^2-4("1")("51")}}{2\times"1"}(=3.48, 14.64)$	ddM1
	$(c=)3.5$ (cm)	A1 [5]
ALT2	M1 M1 as main scheme (then students are working in right angled triangles).	
	$(AX=)10\cos 25$ or $9.(06307787)$	B1 (M1 on ePen)
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25-7\sin("37.14.....")$	ddM1
	or if working with acute angle	
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25+7\sin("37.14.....")$	A1
	$=3.5((cm))$	[5]
		Total 5 marks

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 5 of 10 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
	M1	For a fully correct substitution into the sine rule.
	dM1	For a correct rearrangement to find angle B, implied by sight of the correct acute or obtuse angle. Dependent on previous method mark.
	A1 (M1 on ePen)	For angle C, accept angles which round to 12.
	ddM1	Correct substitution into the cosine rule or sine rule using their angle*s* B and/or C. Allow angle B to be an acute angle and angle C to be the angle that comes from this acute angle using the sum of angles in a triangle. This mark is dependent on the first 2 method marks and is for correct use of the cosine or sine rule, given they have found an angle for B and used this to find angle C.
	A1	For awrt 3.5 (cm)
ALT1	M1	For a fully correct substitution into the cosine rule. Allow AB to be denoted by any letter
	dM1	For a fully correct rearrangement = 0, dependent on the previous method mark.
	A1 (M1 on ePen)	For the correct 3 term quadratic equation = 0.
	ddM1	For a minimally acceptable attempt to solve their quadratic – see general guidance. Dependent on both previous method marks. Sight of 3.48….. or 14.64 may imply this mark
	A1	For awrt 3.5 (cm) Must reject any second solution.
ALT2	M2	As first 2 marks in main scheme
	B1 (M1 on ePen)	For AX either 10cos25 or answers which round to 9
	ddM1	For finding BX correctly and a correct method to find AB, allowing use of acute B
	A1	For awrt 3.5 (cm)

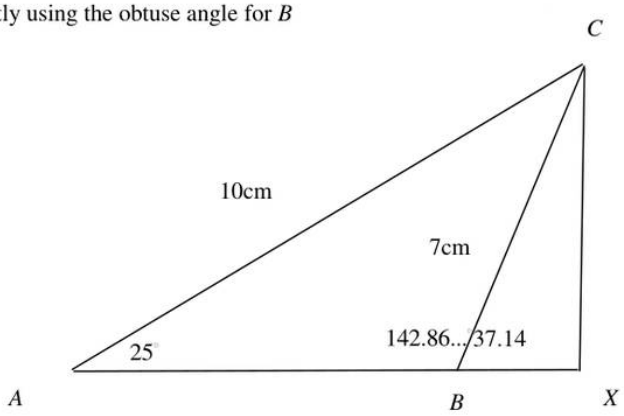
Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

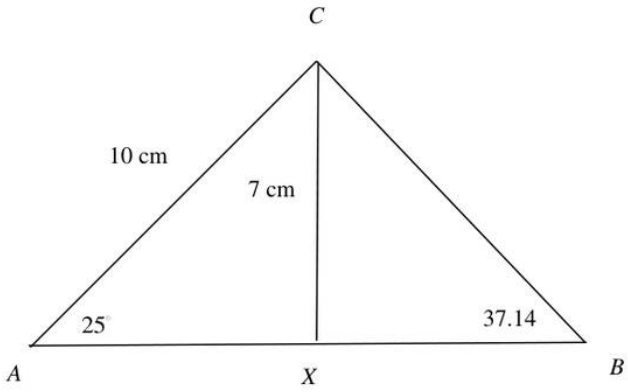
2024 NOV · Paper 01 · extract 6 of 10 · find part c in this mark scheme extract

Particularly to help with ALT2:

If correctly using the obtuse angle for B



Using the acute angle for B



Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 7 of 10 · find part c in this mark scheme extract

Question number	Scheme	Marks
	Remember to look for work on the diagram.	
4	$(FD)=\sqrt{15^2+20^2}$ $(FD=)25$ $(BF=)15\sin 40$ oe $(\Rightarrow (BF)=9.641814145)$ $\sin BDF=\frac{"15\sin 40"}{"\sqrt{15^2+20^2}"}(=0.3856725658)$ or $\sin^{-1}\left(\frac{"9.641814145"}{"25"}\right)$ $(BDF)=22.7°$	M1 A1 M1 M1 dM1 A1 [6]
Example of an ALT	$(DC=)15\times\cos 40$ and $(BD=)\sqrt{20^2+("15\cos 40")^2}$ $(BD=)$awrt 23.1 $(DC=)15\times\cos 40$ and $(FB=)\sqrt{15^2-(15\cos 40)^2}$ oe $(\Rightarrow (FB)=9.641814145)$ $\tan BDF=\frac{"\sqrt{15^2-(15\cos 40)^2}"}{"\sqrt{20^2+("15\cos 40")^2}"}(=0.4180117433)$ or $\tan^{-1}\left(\frac{"9.641814145"}{"23.1"}\right)$ $(BDF)=22.7°$	M1 A1 M1 dM1 M1 A1 [6]
		Total 6 marks

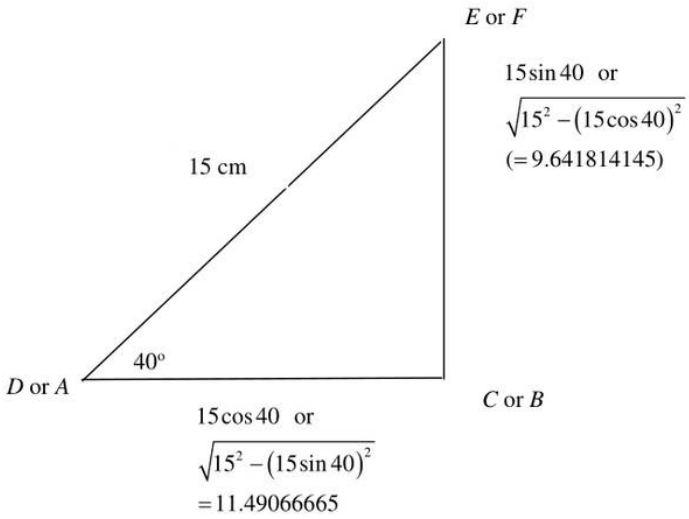
Part	Mark	Additional Guidance
$(BF=)15\sin 40$ oe $(\Rightarrow (BF)=9.641814145)$ means does not need to be labelled *BF* but there mustn't be anything to contradict this. (General principle of marking) It is fine to label with equivalent sides eg (*EC* = *FB*) There are a number of ways candidates are completing this question. Please mark to the following general principles – send anything to review that can't be marked using these.		
	M1	For a **full** and correct method to find *FD* or *BD*
	A1	For *FD* = 25 or *BD* = awrt 23.1
	M1	For a **full** and correct method to find a second side in triangle *BDF*
	M1	For correctly identifying the correct angle to be found. Dependent on previous method mark. Look for any clear identification that this is the angle they're finding, **including on the diagram** or in written work.
	dM1	For the use of their values for two sides in triangle *BDF* in a correct trig **equation** or the correct **expression** to find the angle with the relevant inverse trig function. There must be a full, complete and correct **method** which leads to an equation or expression to find angle *BDF*, part methods should not be awarded this mark. Dependent on the previous method mark only ie the correct angle identified, which may be implied in a correct equation or expression. If it is clear candidates are using 2 of their values for *BF*, *FD* and *BD*, this mark can be awarded, even if the work to calculate either is incorrect. $\sin BDF=\left(\frac{"BF"}{"DF"}\right)$ or $\cos BDF=\left(\frac{"BD"}{"DF"}\right)$ or $\tan BDF=\left(\frac{"BF"}{"BD"}\right)$ or $\sin^{-1}\left(\frac{"BF"}{"DF"}\right)$ or $\cos^{-1}\left(\frac{"BD"}{"DF"}\right)$ or $\tan^{-1}\left(\frac{"BF"}{"BD"}\right)$ If incorrect lengths are found, you may allow $\left(\frac{"BF"}{"DF"}\right)$ or $\left(\frac{"BD"}{"DF"}\right)>1$ for this mark Note: if students choose to use cosine rule, they must additionally use a full and correct method to find a third side of triangle *BDF* and have either of these $\cos BDF=\frac{DF^2+BD^2-BF^2}{2\times DF\times BD}$ or $\cos^{-1}\left(\frac{DF^2+BD^2-BF^2}{2\times DF\times BD}\right)$
	A1	Any answer which rounds to (awrt) 22.7°

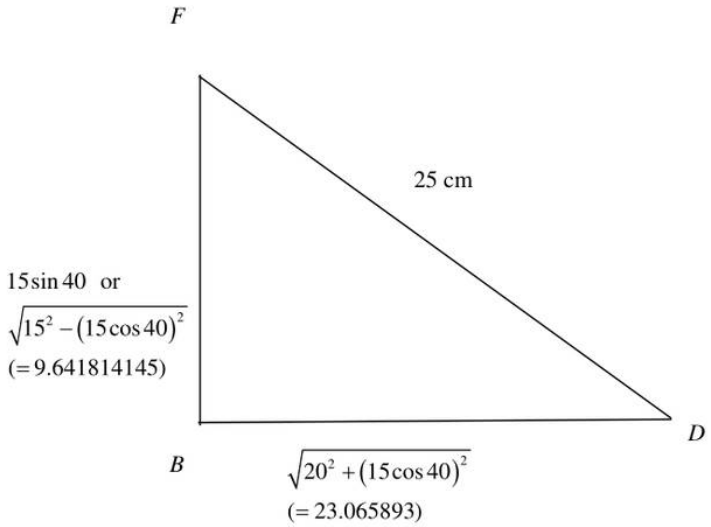
Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 8 of 10 · find part c in this mark scheme extract

Useful triangles and sides.





Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 9 of 10 · find part c in this mark scheme extract

Question number	Scheme	Marks
5 (a)	$3t^2-16t+5=0 \rightarrow (3t-1)(t-5) \rightarrow t =$ $t=\frac{1}{3}$ oe $\quad t=5$	M1 M1 A1 [3]
(b)	$(a=)6t-16>0$ $t>\frac{8}{3}$ oe	M1 A1 [2]
(c)	$\int_{(1)}^{(4)}(3t^2-16t+5)\,dt$ $\left[t^3-8t^2+5t\right]_{(1)}^{(4)}$ $\left(4^3-8(4)^2+5(4)\right)-\left(1^3-8(1)^2+5(1)\right)$ -42 (distance) = 42	 M1 A1 M1 A1 A1 [5]
		Total 10 marks

Part	Mark	Additional Guidance
(a)	M1	For setting $v = 0$ and attempting to solve, see general guidance for definition of a minimum attempt to solve leading to $t =$
	M1	For a correct value of t. Any correct value will imply M2.
	A1	For both correct values of t Both correct values will imply M2 A1
(b)	M1	For an attempt to differentiate and setting their expression > 0 (allow placing = 0 as a concession) See general guidance for definition of a valid attempt. Additionally for this mark, no power of t to increase, at least one term differentiated correctly.
	A1	For $t>\frac{8}{3}$ oe Accept rounding to 1 dp or better eg 2.7, 2.67, 2.667 etc Or indication of recurring e.g. $2.\dot{6}$ or $2.6'$ or 2.6... minimum 3 dots
(c)	M1	For an attempt to integrate the given expression for v, see general guidance for the definition of a valid attempt. Additionally, no power of t to decrease. Limits do not need to be present.
	A1	For a correct integration. Limits do not need to be present.
	M1	For correct substitution of limits into **their** changed expression. Can be any changed expression with a minimum of 2 terms. This mark may be implied by sight of – 42 or 42 or – 44 – – 2 or 44 – 2. If the final answer is incorrect, must see substitution of both limits correctly at least once. For this mark, students may assume an arbitrary constant (typically $C = 0$) or leave an arbitrary constant in their working and have the equivalent $\left(4^3-8(4)^2+5(4)+C\right)-\left(1^3-8(1)^2+5(1)+C\right)$ Candidates cannot attain this mark for just – 44 and – 2 without a subtraction.
	A1	For – 42 or 42.
	A1	Correct positive distance given.
Any candidate not showing the step of integration – 0 marks for this part of the question.		

Type MS · Finding n from an equation linking sums and terms

Answer 8 of 11 · Question 2 part c · 4 marks

2024 NOV · Paper 01 · extract 10 of 10 · find part c in this mark scheme extract

Question number	Scheme	Marks
6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$ $=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	M1 A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe $\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors $\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3$ $P=\frac{a}{2}, Q=-\frac{ab}{16}*, R=\frac{3ab^2}{256}, S=-\frac{5ab^3}{2048}*$	M1 M1 A1 A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$ $\left(b^2=1\right)b=1*$	M1 A1*

Type MS · Finding n from an equation linking sums and terms

Answer 9 of 11 · Question 5 part c · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

5(a)	$\sum_{r=1}^{n}(3r+5)=8+11+14+...+(3n+5)=\frac{1}{2}n(8+3n+5)=\frac{1}{2}n(3n+13)$ *	M1M1A1cso (3)
ALT	$\sum_{r=1}^{n}(3r+5)=\sum_{1}^{n}3r+5n=\frac{n}{2}(3+3n)+5n=\frac{n}{2}(13+3n)$ *	M1M1A1cso
(b)	$\sum_{r=35}^{50}(3r+5)=\frac{50}{2}(13+150)-\frac{34}{2}(13+102)$	M1
	$=2120$	A1 (2)
(c)	$\frac{n}{2}(13+3n)=385$	
	$3n^2+13n-770=0$	M1
	$(3n+55)(n-14)=0 \quad n=14$	M1A1 (3)
		[8]

(a)	
M1	Evaluate **either** first and last terms **or** first and common difference
M1	Use either sum formula. Can be shown explicitly or implied by a correct, full substitution of n and their a and their d or l
A1cso	Reach the **given** result with no errors in the working. Must be the complete result, not just the RHS or there must be a conclusion eg "shown"
ALT	
M1	Split the $(3r + 5)$ into 2 parts and deal with the 5 correctly
M1	Sum $(3r)$ either by using a summation formula or by using the standard result
A1	Reach the **given** result with no errors in the working
(b)	
M1	Express the required sum as the difference of 2 sums. Second sum must have 34 terms. Use the result given in (a). **Using a standard formula** with first term and either last term or common difference scores 0/2 as question states "hence". Calculator solutions (without showing the difference of the 2 sums first) score M0
A1	Correct answer.
(c)	
M1	Use the result in (a) or some other valid method to form a 3 term quadratic in n
M1	Solve their 3TQ by any valid means. Must reach n = ... Negative value need not be seen
A1	Correct answer. n = 14 and no other Correct quadratic followed by correct answer scores 3/3 Incorrect quadratic solved by calculator M0A0

Type MS · Finding n from an equation linking sums and terms

Answer 9 of 11 · Question 5 part c · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$-3 \quad -5$	

Type MS · Finding n from an equation linking sums and terms

Answer 10 of 11 · Question 3 part c · 3 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

3(a)	$\left(\sum_{r=1}^{n}(5r-3)\Rightarrow\right) a=2,\quad d=5$	B1
	$\left(\sum_{r=1}^{n}(5r-3)=\right)\frac{n}{2}(2\times2+(n-1)5)$ or $\frac{n}{2}(4+5n-5)$	M1
	or $\frac{n}{2}(2+2+(n-1)5)$ or $\frac{n}{2}(2+5n-3)$	
	$=\frac{n}{2}(5n-1)*$	A1 cso [3]
	ALT (Using standard results)	
	$\left(\sum_{r=1}^{n}(5r-3)=\right)5\sum_{1}^{n}r-3\sum_{1}^{n}1$	B1
	$\left(\sum_{r=1}^{n}(5r-3)=\right)5\left[\frac{n}{2}(n+1)\right]-3n=\frac{5n^2+5n-6n}{2}=\frac{n}{2}(5n-1)*$	M1A1 cso [3]
(b)	$\left(\sum_{31}^{60}(5r-3)=\right)\frac{60}{2}(5\times60-1)-\frac{30}{2}(5\times30-1)=6735$	M1A1 [2]
	ALT	
	$\left(\sum_{31}^{60}(5r-3)=\right)\frac{30}{2}(152+297)=6735$	M1A1 [2]
	or $\left(\sum_{1}^{60}(5r-3)-\sum_{1}^{30}(5r-3)=\right)\frac{60}{2}(2\times2+(60-1)\times5)-\frac{30}{2}(2\times2+(30-1)\times5)=6735$	
(c)	$\frac{n}{2}(5n-1)=3783\Rightarrow5n^2-n-7566=0$	M1
	$\Rightarrow(5n+194)(n-39)=0$	M1
	$\Rightarrow n=39\quad\left[-\frac{194}{5}\right]$	A1 [3]
		Total 8 marks

Type MS · Finding n from an equation linking sums and terms

Answer 10 of 11 · Question 3 part c · 3 marks

2024 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the values of a and d, these may be explicitly stated or implicitly used in a formula.
	M1	Correctly substitutes their values of a and d into the correct summation formula
	A1 cso	For the correct expression, minimum steps as shown, no errors or omissions.
	ALT	
	B1	For writing the given expression as $5\sum_{1}^{n} r - 3\sum_{1}^{n} 1$
	M1	For the sum shown, using standard results for the series.
	A1 cso	For the correct expression, minimum steps as shown, no errors or omissions.
(b)	M1	Uses the given expression with both $n = 60$ and $n = 30$ and subtracts. Indicated by 8970 – 2235. As a concession, allow substitution of $n = 31$ and 60 for the mark. **ALT** Correct substitution of the correct 'first' and 'last' value into the correct summation formula ie $\frac{30}{2}(152+297)$. Allow as a concession $\frac{29}{2}(152+297)$ or correct substitution of both $n = 30$ and $n = 60$ into the correct sum to n terms formula and a subtraction. Indicated by 8970 – 2235. As a concession, allow substitution of $n = 31$ and 60 for the mark.
	A1	For 6735
(c)	M1	For correctly placing the given expression = 3783 and rearranging (allow one error) to get a 3TQ = 0.
	M1	For solving their 3TQ and achieving at least one value. Minimum attempt to solve required (see general guidance). A correct value of 39 can imply this mark
	A1	For 39 only. If $-\frac{194}{5}$ is also given as a solution, withhold this mark.

Type MS · Finding n from an equation linking sums and terms

Answer 10 of 11 · Question 3 part c · 3 marks

2024 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding n from an equation linking sums and terms

Answer 11 of 11 · Question 5 part c · 4 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

5(a)	**Mark both parts together**	
	$4k+1=a+(6-1)d \Rightarrow [\ 4k+1=a+(6-1)d\ ]$	M1
	$36k+1=\frac{10}{2}(2a+9d) \Rightarrow [36k+1=10a+45d]$	M1
	Solves the simultaneous equations by any method:	M1
(i)	$d=\frac{4k+9}{5}$	A1
(ii)	$a=-8$*	A1 cso [5]
(b)	$7=-8+3d \Rightarrow 7=-8+3\left(\frac{4k+9}{5}\right) \Rightarrow 25=4k+9 \Rightarrow k=4$ *	M1A1 cso [2]
(c)	$\left[d=\frac{4\times4+9}{5}=5\right]$ you may see this in part (c)	
	$\frac{n}{2}(2\times-8+(n-1)\times5)=5[-8+(n+10-1)5]+105$	M1
	$\Rightarrow -21n+5n^2=370+50n+210 \Rightarrow 5n^2-71n-580=0$	dM1
	$\Rightarrow (5n+29)(n-20)=0 \Rightarrow n=20$	M1A1 [4]
		Total 11 marks